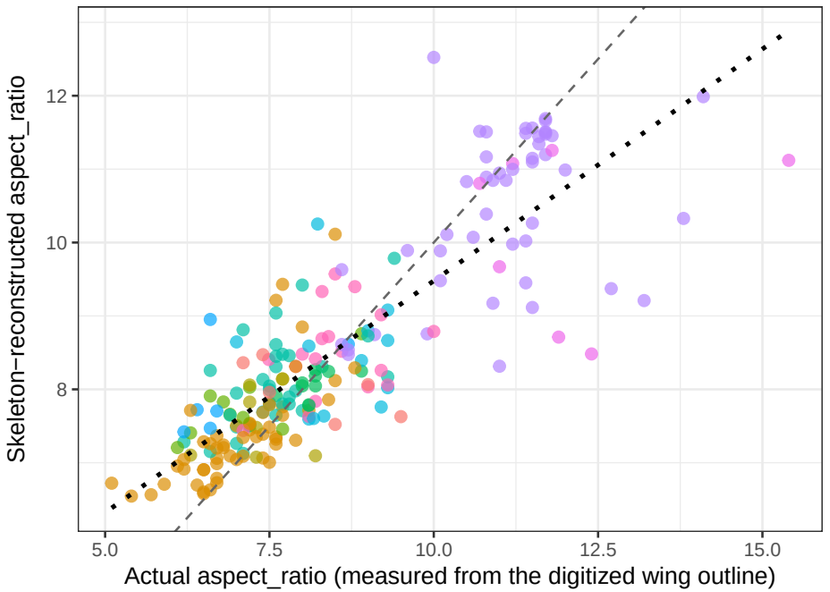
Skeleton−only reconstruction vs. actual (curved−wing) aerodynamic properties

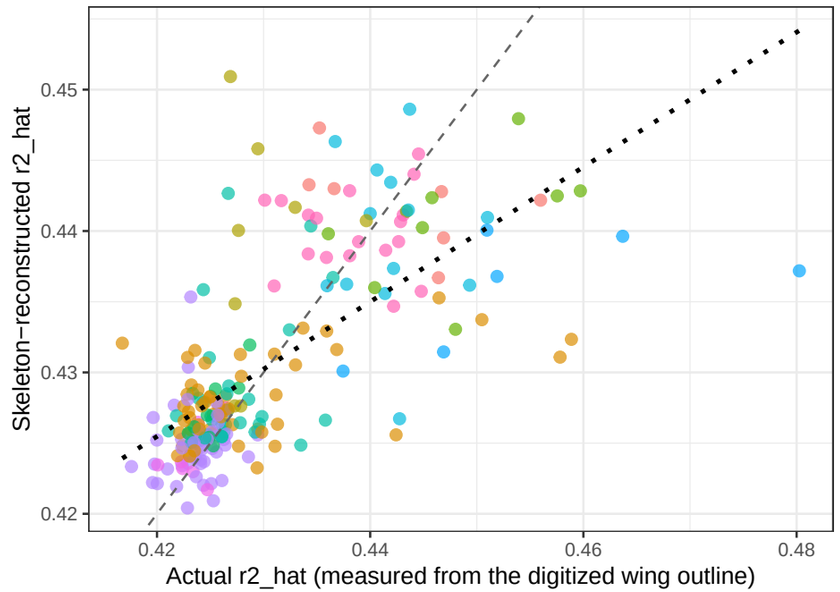
aspect_ratio

$R^2 = 0.71$ (skeleton−only OOB reconstruction vs. outline−measured value)



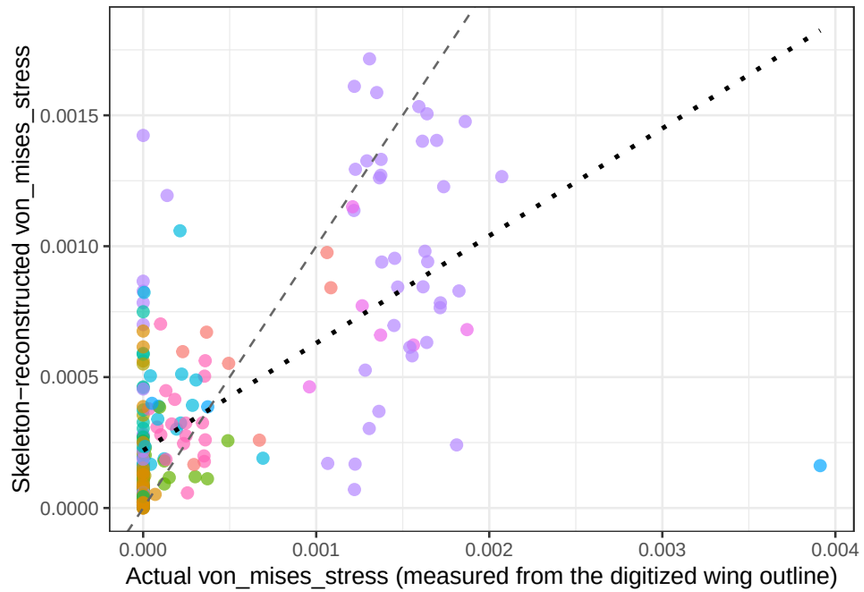
r2_hat

$R^2 = 0.463$ (skeleton−only OOB reconstruction vs. outline−measured value)



von_mises_stress

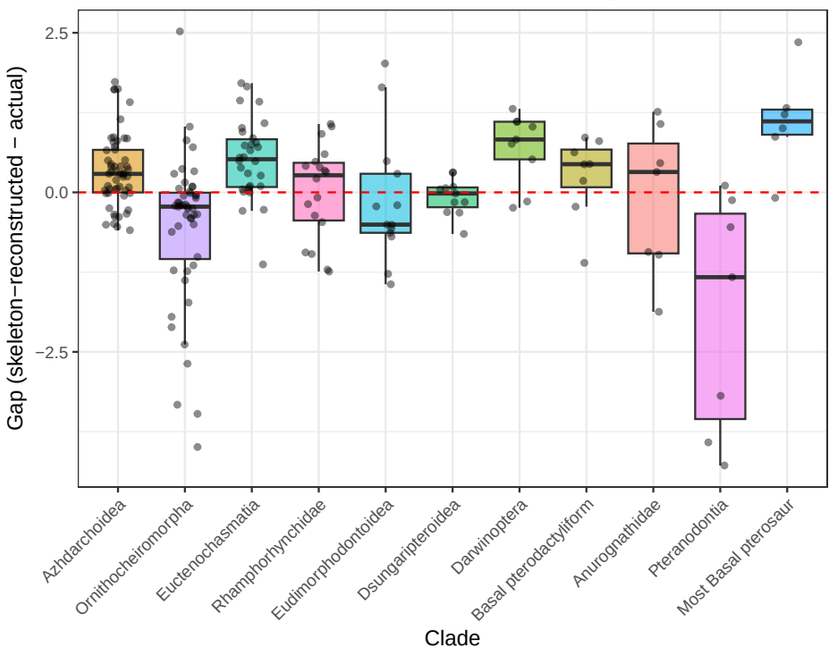
$R^2 = 0.409$ (skeleton−only OOB reconstruction vs. outline−measured valu



clade
Anurognathidae
Azhdarchoidea
Darwinoptera
Dsungaripteroidea
Eudimorphodontoidea
Most Basal pterosaur
Pteranodontia
Rhamphorhynchidae
1:1 (skeletal based)
Fitted trend
Basal pterodactyliform
Euctenochasmatia
Ornithocheiromorpha

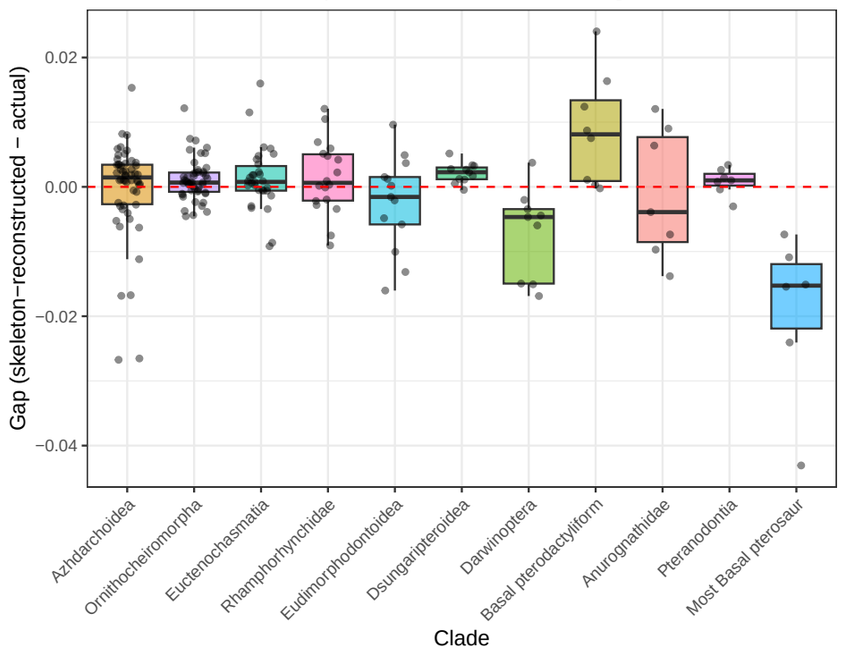
aspect_ratio gap by clade

Positive = skeleton over−predicts relative to the real wing outline



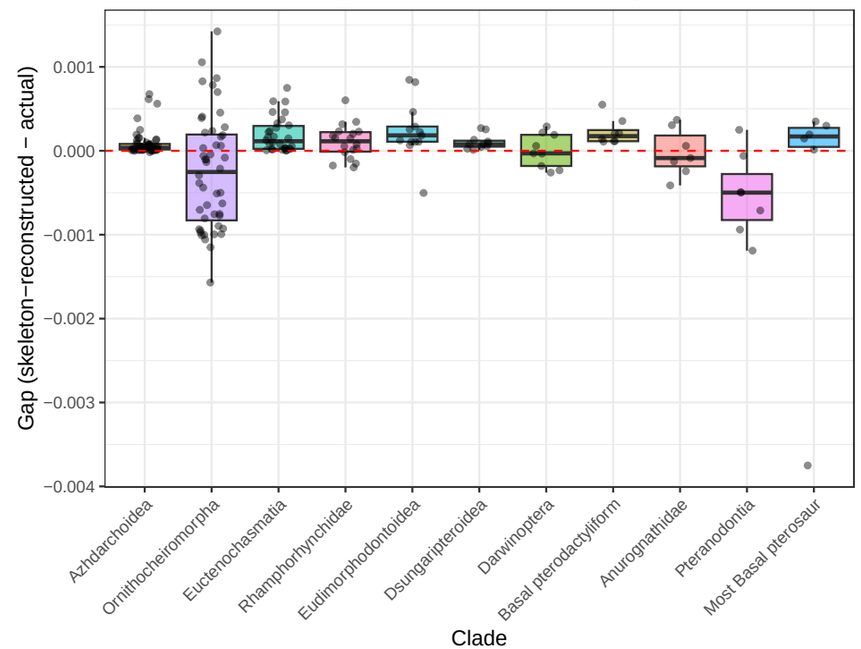
r2_hat gap by clade

Positive = skeleton over−predicts relative to the real wing outline



von_mises_stress gap by clade

Positive = skeleton over−predicts relative to the real wing outline



rom the real, digitized wing outline; dashed = 1:1 perfect reconstruction, dotted = fitted trend. Bottom: reconstructed−minus−actual gap per clade; a positive gap means the skeleton−only reconstruction overestimates performance relative to the real, digitized wing outline.